Microsoft 365

MD-100 Windows Client

Module 05: Configure storage on Windows clients

Module Agenda

- Manage storage on Windows clients
- Maintain disks and volumes
- Implement Storage Spaces

Lesson01: Manage storage on Windows clients

Lesson 1: Manage storage on Windows clients

- Understand storage options
- Examine network and cloud storage options
- Describe MBR and GPT paritions
- Describe dynamic disks
- Explore disk management tools
- Describe simple volumes
- Compare mirrored, spanned and striped volumes
- Manage existing volumes

Understand storage options

Local hard disk:

- SSD
- Hard disk drive

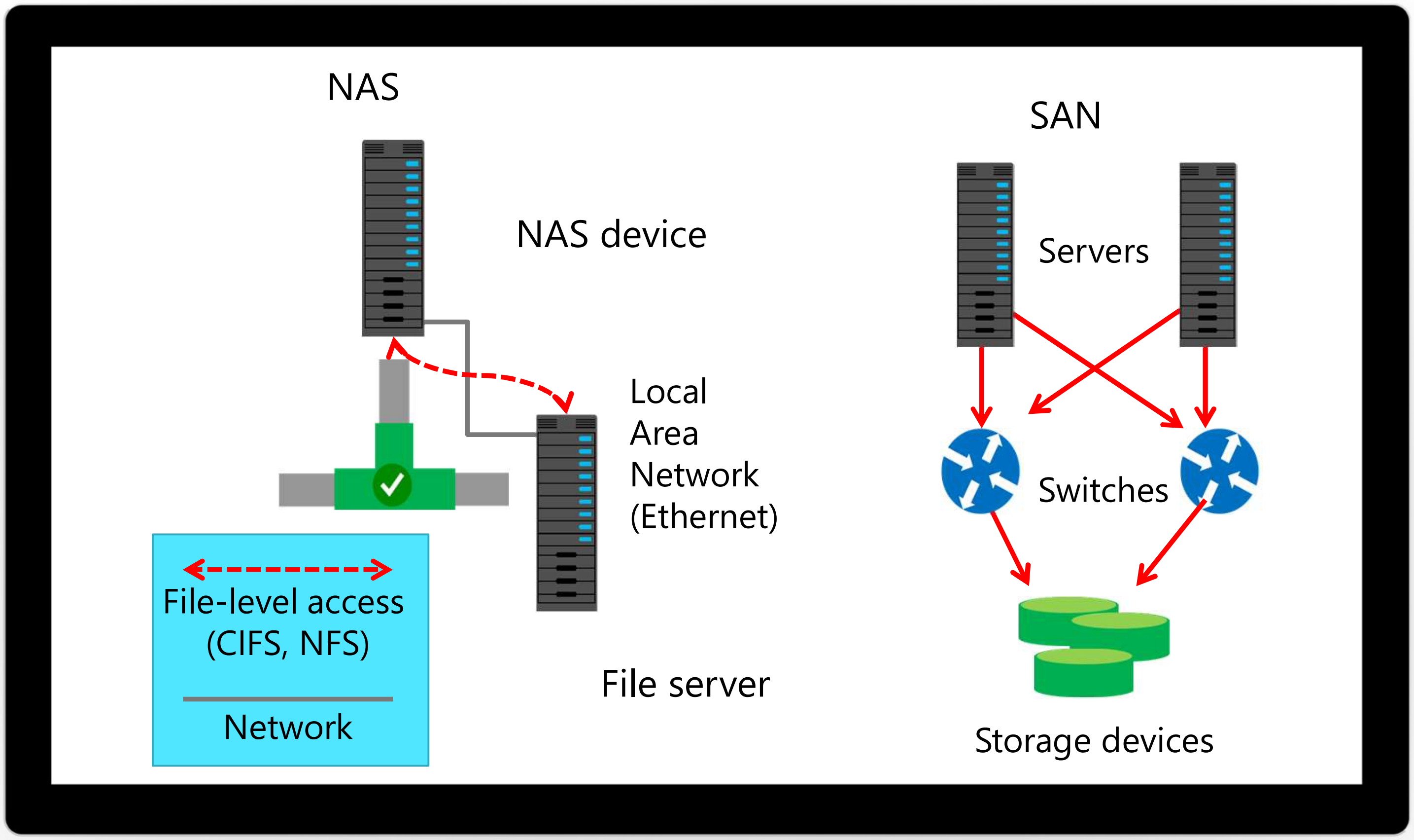
Virtual hard disk:

- .vhd (up to 2 TB)
- .vhdx (up to 64 TB)

Server-based storage:

- File server
- NAS
- SAN

Examine network and cloud storage options



Describe MBR and GPT paritions

MBR disk:

- Contains the partition table for the disk and a small amount of executable code called the master boot code
- Is on the first sector of the hard disk and is created when a disk is partitioned
- Maximum four partitions of 2 TB each

GPT disk:

- Contains an array of partition entries describing the start and end LBA of each partition on a disk
- Supports up to 128 partitions and a theoretical 18 exabyte size
- Enhances reliability
- Supports boot disks on 64-bit Windows operating systems and UEFI systems

Describe dynamic disks

Consists of multidisk volumes:

- Spanned
- Striped
- Mirrored

Can contain up to 1024 volumes

Can convert from basic disk without data loss

Requires all volumes to be deleted when converting to basic disk

Can be managed by using DiskPart or Disk Management:

- There are no Windows PowerShell cmdlets for managing dynamic disks

Explore disk management tools

Disk Management snap-in:

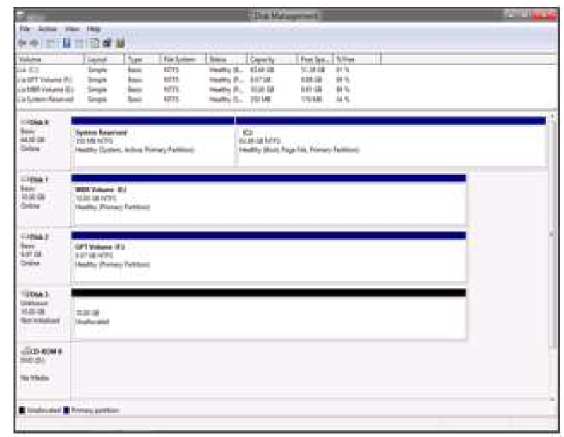
- GUI
- Manage disks and volumes, both basic and dynamic,locally or on remote computers
- Simple partition creation

DiskPart:

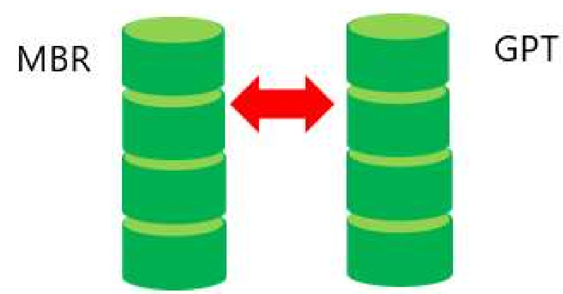
- Scriptable command-line tool
- Create scripts to automate disk-related tasks
- Always runs locally

Windows PowerShell:

- Has native disk management commands
- Can be used to script disk-related tasks



Use DiskPart or Windows PowerShell to convert partition styles

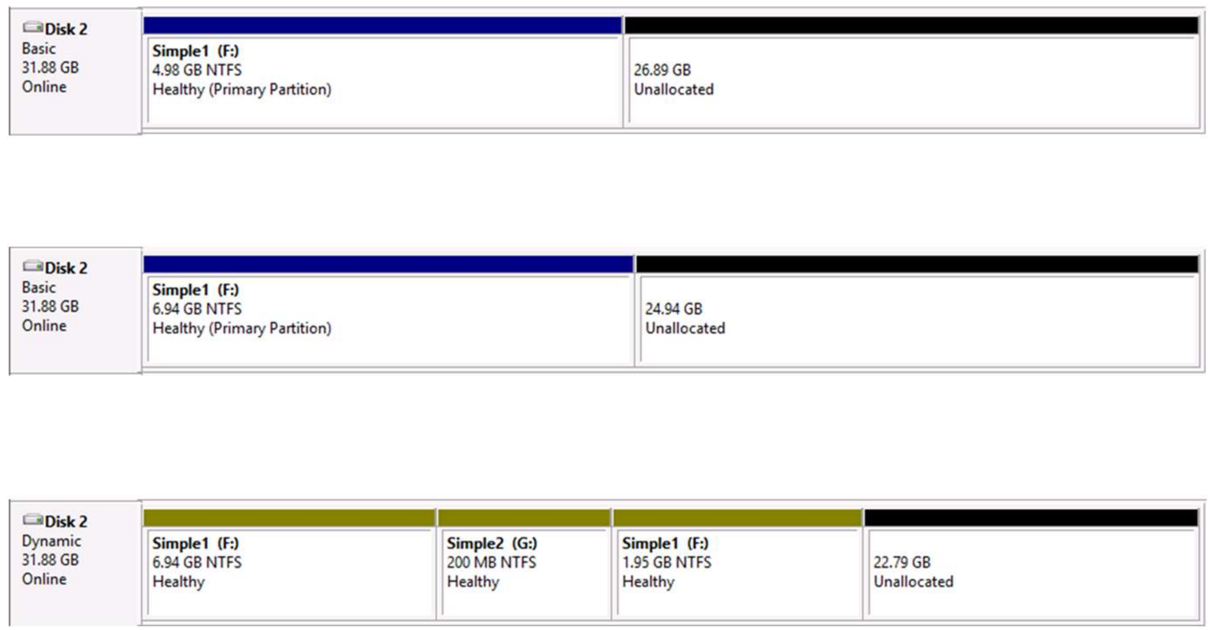


Describe simple volumes

Simple volume that encompasses available free space from a single, basic, or dynamic hard disk drive

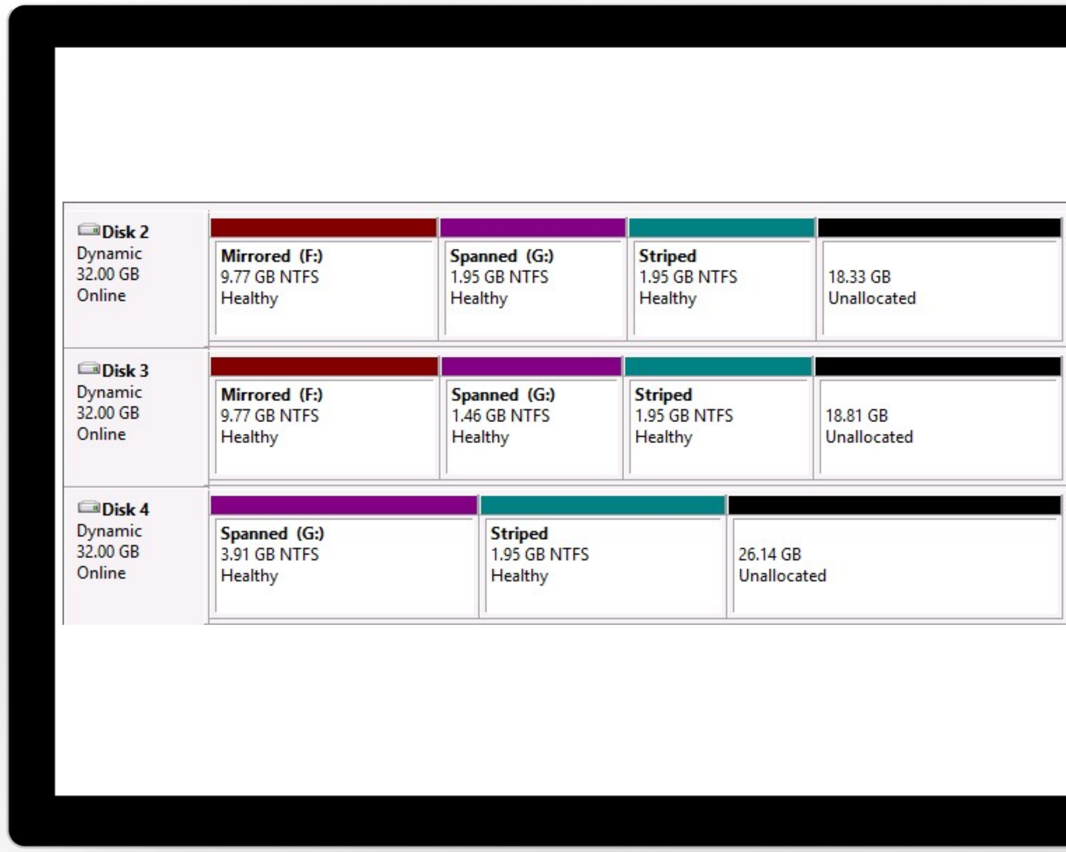
Can be extended if contiguous space is free on the same disk

If extending into noncontiguous space, the disk will be converted to dynamic if it is a basic disk



Compare mirrored, spanned and striped volumes

- Join areas of unallocated space on disks into a single logical disk
- Mirrored:
  - Disk space is allocated once and used simultaneously
- Spanned:
  - Disk space is added and used sequentially
- Striped:
  - Disk space is allocated once and used equally across every physical disk in the striped set



Manage existing volumes

- Resize a volume to create additional, unallocated space to use for data or apps on a new volume.
- Shrink simple and spanned dynamic disks to:
  - Extend a simple volume on the same disk
  - Extend a simple volume to include unallocated space on other disks on the same computer
- Before shrinking:
  - Defragment the disk
  - Ensure that the volume you want to shrink does not contain any page files

Knowledge Check

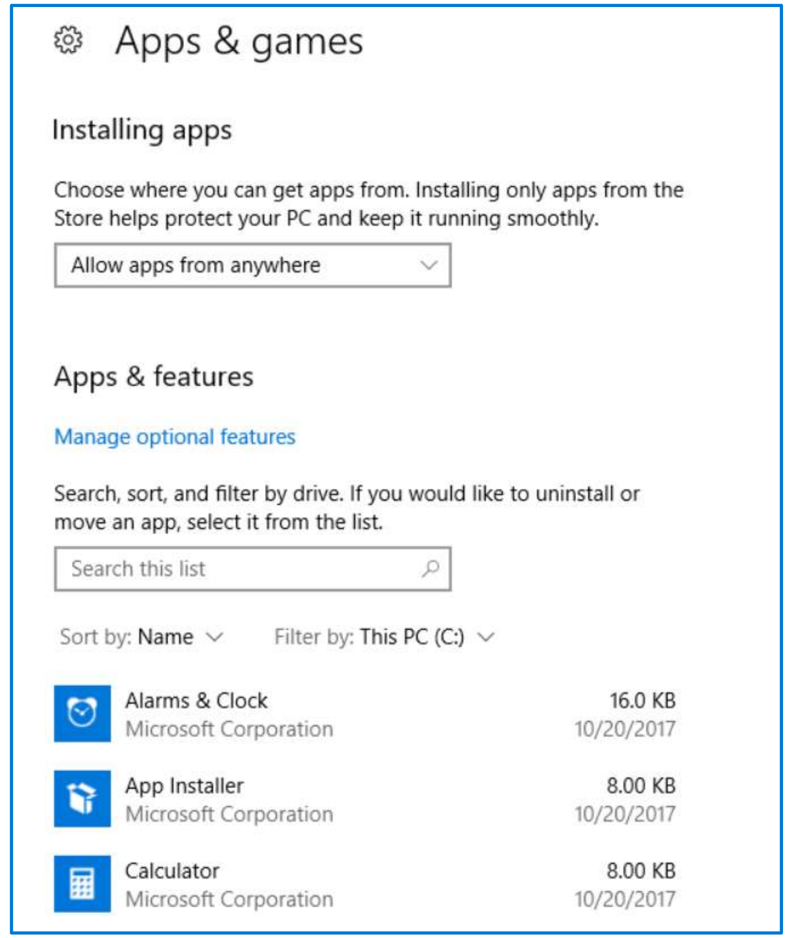
Lesson02: Maintain disks and volumes

Lesson 2: Maintain disks and volumes

- Monitor storage usage
- Understand disk optimization
- Explore file and folder compression
- What are disk quotas?

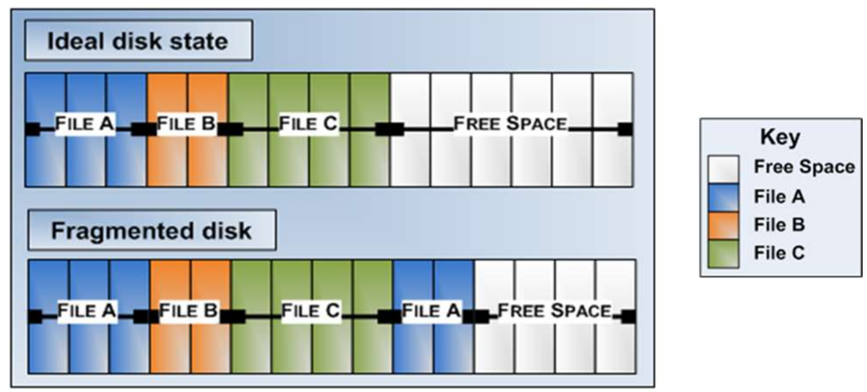
Monitor storage usage

- Presents an overview of storage usage by:
  - Drive (internal, external, and OneDrive)
  - 13 categories including System, Apps, Music, and Pictures
- Enables you to choose the drive to which you want to save new files, such as:
  - Apps
  - Music
  - Documents
  - Videos
  - Pictures
- Storage Sense can automatically free up space



Understand disk optimization

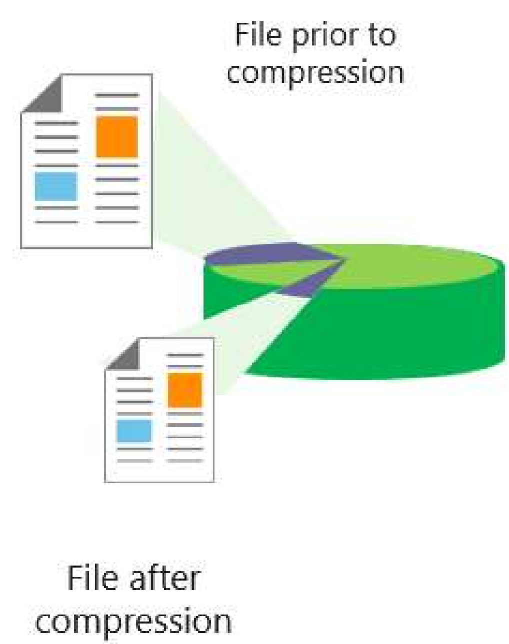
- Defragmentation not needed on SSDs.
  - Windows Auto-Detects Disk Type
  - Optimize will run TRIM on SSDs
- Disk fragmentation on HDDs can:
  - Consist of both fragmented files and fragmented free space
  - Lead to poor performance of a disk subsystem
- Default settings are usually sufficient.



Explore file and folder compression

The NTFS file system uses NTFS file compression to compress files, folders, and volumes:

- Uses compression to save disk space
- Does not use compression for system files and folders
- Compression is configured as an NTFS attribute
- NTFS calculates disk space based on uncompressed file size
- Applications that open a compressed file only see the uncompressed data



Knowledge Check

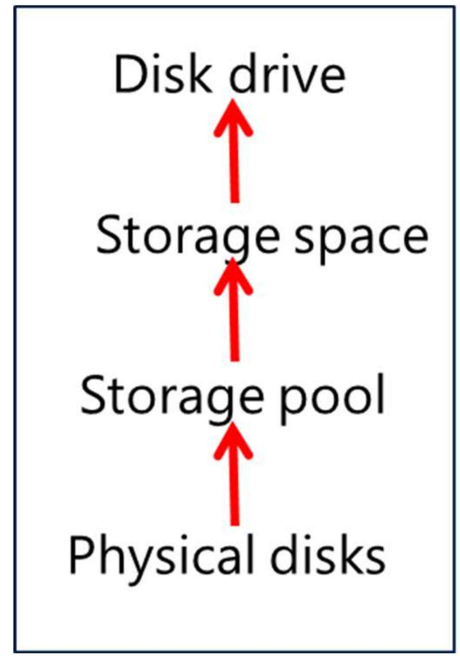
Lesson03: Implement Storage Spaces

Lesson 3: Implement Storage Spaces

- Explore Storage Spaces
- Examine features of Storage Spaces
- Examine scenarios for Storage Spaces

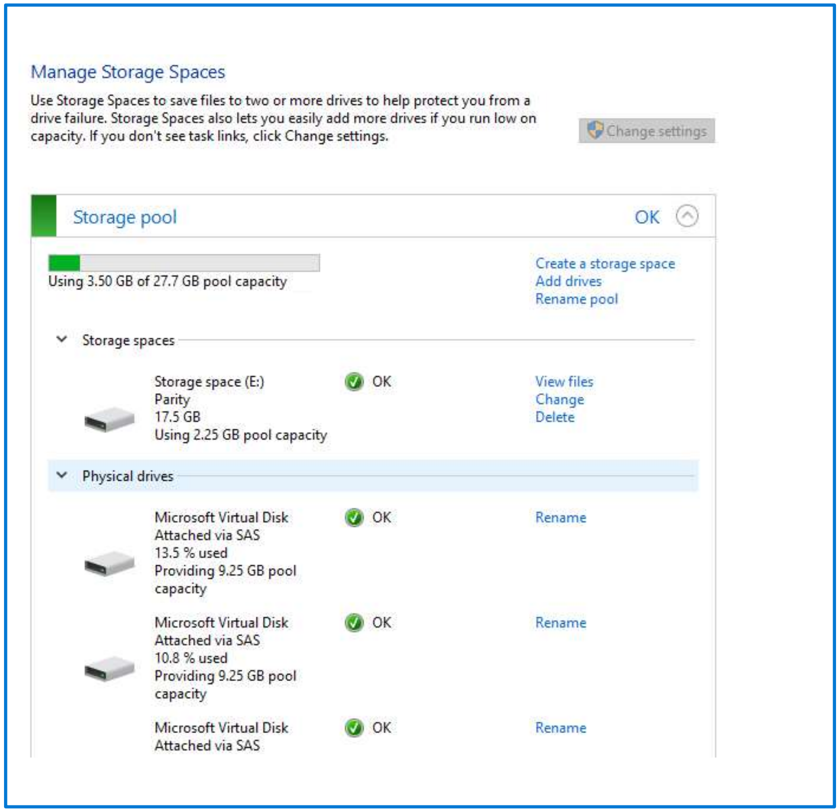
Explore Storage Spaces

- Use Storage Spaces to add physical disks of any type and size to a storage pool, and then create highly available virtual disks from the storage pool
- To create a storage space, you need the following:
  - One or more physical disks
  - Storage pool that includes the disks
  - Storage space that is created with disks from the storage pool
  - Disk drives that are based on storage spaces



Examine features of Storage Spaces

Feature	Options
Storage layout	• Simple • Two-way or three-way mirror • Parity
Provisioning schemes	• Thin vs. fixed provisioning



Examine scenarios for Storage Spaces

- Thin provisioning scenario:
  - Easier storage growth
  - Add disks when the need arises
- Reliable storage scenario:
  - Fault tolerance
  - No data loss
- High-performance scenario:
  - Parity resilience gives better performance with SSDs
  - Usable for video editing and other high disk I/O scenarios

Knowledge Check

Practice Labs

Managing Storage

Creating a Storage Space

Module Review

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